DEMO / SYNTHETIC: Wire feed rate and travel speed effects on WAAM Inconel 718 walls

Authors: Demo Author E
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Abstract

DEMO / SYNTHETIC TEXT generated to test the AM-KG pipeline; the numbers are not real research results. Inconel 718 thin walls were deposited by wire arc additive manufacturing (WAAM). The wire feed rate and travel speed affected the dendritic structure. Laves phase formation increased hot cracking.

2. Methodology

IN718 thin walls were built by WAAM with a wire feed rate of 6 m/min, a travel speed of 5 mm/s and an arc current of 160 A. The interlayer temperature was kept below 150 °C. Microstructures were analysed using EBSD and SEM, and tensile tests were carried out on machined specimens.

3. Results and discussion

Increasing the travel speed reduced the heat input and refined the dendritic structure. Coarse columnar grains were observed along the build direction. The Laves phase increased hot cracking in the interdendritic regions. The tensile strength of the as-deposited walls was 820 MPa. Solution treatment and aging dissolved the Laves phase and improved the tensile strength to 1150 MPa.

4. Conclusions

Travel speed influenced the grain size of WAAM Inconel 718.